Radiator fin array Specification

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Revision	C
Status	APPROVED
Classification	INTERNAL
Author	Dalric Torvald
Approved by	Aldric Voss
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1. Purpose

This specification defines the performance, interface and verification requirements for the radiator fin array (TX-RFA) of the Thermal Control & Heat Rejection on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-TX-03 (WBS 1.2.3), owned by Jessa Pell.

The prime supplier for flight hardware is Lianna Orbital Systems. Responsible engineer: Dalric Torvald.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-TX-DWG-0202	Radiator fin array general arrangement drawing
SHV-TX-BOM-0302	Radiator fin array bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-HR-TX	Interface Control Document: Hypermatter Reactor / Thermal Control & Heat Rejection

4. Description

Thermal control carries surplus reactor and habitat heat to the radiator fin arrays on the station surface through heat exchanger stacks, pumped coolant loops and passive heat pipes.

The radiator fin array is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
TX-RFA-R01	The radiator fin array shall provide a fin area of 6.3 km².	6.3 km²	±2%	Demonstration
TX-RFA-R02	The radiator fin array shall provide a emissivity of 0.89.	0.89	±5%	Demonstration

5.1 General requirements

ID	Requirement	Verification
TX-RFA-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
TX-RFA-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
TX-RFA-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
TX-RFA-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
TX-RFA-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the TX test-report series.

8. Revision history

Rev	Date	Description	Author
A	2023-12-01	Initial release.	Dalric Torvald
B	2024-11-01	Incorporated interface changes.	Dalric Torvald
C	2025-03-21	Fin area set to 6.3 km² (approved).	Dalric Torvald